

AN ELEMENTARY PROOF OF ASYMPTOTIC NORMALITY FOR LINEAR RANK STATISTICS

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Summary: A well-known result in nonparametric statistical theory is that the large-sample distribution of a linear rank statistic is approximately normal under certain regularity conditions. Existing proofs of this in the case of i.i.d. observations, see e.g. Hájek & Šidák (1967, chapter V) or Randles & Wolfe (1979, chapter 8), require the use at one point or another of technical results which have no obvious statistical or probabilistic content. The purpose of this note is to outline a simple method of proof which for the most part avoids such technicalities.

1. INTRODUCTION AND NOTATION

Let $U_1, \dots, U_n$ be independent uniform (0,1) random variables with order statistics $U_{1,n} < \dots < U_{n,n}$ and ranks $R_{1n}, \dots, R_{nn}$: $U_i = U_{R_{in},n}$.

Consider a simple linear rank statistic

$$(1) \quad \sum_{i=1}^n c_i a_n(R_{in})$$

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where $\{c_{in}; 1 \leq i \leq n\}$ are constants with

$$\sum_{i=1}^n c_{in} = 0, \quad \sum_{i=1}^n c_{in}^2 = 1, \quad n \geq 2; \quad \max_{1 \leq i \leq n} |c_{in}| \rightarrow 0$$

and $\{a_n(i); 1 \leq i \leq n\}$ are scores determined by a square integrable function φ on $(0,1)$, either by

$$(2) \quad a_n(i) = E\varphi(U_{i,n})$$

or by

$$(3) \quad a_n(i) = \varphi(i/(n+1)).$$

Let $T_{n,2}$ and $T_{n,3}$ denote expression (1) when the scores are given by (2) and (3), respectively.

Under the stated conditions on the c_{in} the asymptotic normality of

$$T_{n,1} = \sum_{i=1}^n c_{in} \varphi(U_i)$$

follows easily from Lindeberg's theorem and asymptotic normality of $T_{n,j}$ for $j = 2, 3$ then follows from the relations

$$(4) \quad E(T_{n,1} - T_{n,j})^2 \rightarrow 0; \quad j = 2, 3.$$

Existing proofs of (4) all seem to require some rather technical auxiliary results, see e.g. Hájek & Šidák (1967, section V.1.4) or Randles & Wolfe (1979, section 8.3). Lehmann (1975, appendix 4) gives a nice proof for *bounded* scores. Many interesting scores, for instance the Van der Waerden scores or the exponential ordered (Savage) scores are, however, unbounded. In the sequel we assume that φ , in addition to being square integrable, is expressible as a difference of two nondecreasing left continuous functions. The added restriction on φ is of no practical consequence since all

commonly used score functions conform to it. However, it results in a substantial gain in theoretical simplicity and leads to an elementary proof of equation (4).

2. RESULTS

Our point of departure is the well-known relation (Hájek & Šidák (1967, page 157) or Randles & Wolfe (1979, page 288))

$$(5) \quad T_{n,2} = E(T_{n,1} | R_{1n}, \dots, R_{nn})$$

which implies, by a standard property of conditional expectation, that

$$E(T_{n,1} - T_{n,2})^2 = E(T_{n,1})^2 - E(T_{n,2})^2.$$

Since

$$(6) \quad E(T_{n,1})^2 = \int_0^1 \varphi^2(u) du - \left(\int_0^1 \varphi(u) du \right)^2 \equiv A^2.$$

the relation (4) will then follow at once for $j = 2$ if we prove

$$(7) \quad E(T_{n,2})^2 \rightarrow A^2$$

and for $j = 3$ if we prove

$$(8) \quad E(T_{n,2} - T_{n,3})^2 \rightarrow 0.$$

Let $I(\cdot)$ denote the indicator function and for $0 < v < 1$ set

$$T_{n,1}(v) = \sum_{i=1}^n c_{in} \{I(U_i \leq v) - v\}$$

$$T_{n,3}(v) = \sum_{i=1}^n c_{in} \{I(R_{in} \leq (n+1)v) - v\}$$

and

$$T_{n,2}(v) = \sum_{i=1}^n c_{in} \{b_n(R_{in}, v) - v\}$$

with

$$(9) \quad b_n(i, v) = \Pr(U_{in} \leq v) = E(I(U_j \leq v) | R_{jn} = i).$$

Since both $T_{n,1}(v)$ and $T_{n,3}(v)$ are zero in neighbourhoods of $v = 0$ and $v = 1$ and since $T_{n,j} = \int_0^1 \varphi(v) dT_{n,j}(v)$ for $j = 1, 3$, an integration by parts shows that

$$(10) \quad T_{n,1} = - \int_0^1 T_{n,1}(v) d\varphi(v)$$

and

$$(11) \quad T_{n,3} = - \int_0^1 T_{n,3}(v) d\varphi(v).$$

Conditioning both sides of (10) on $(R_{1n}, \dots, R_{nn})$ and using (5) and (9) we also get

$$(12) \quad T_{n,2} = - \int_0^1 T_{n,2}(v) d\varphi(v).$$

The assumption on φ ensures the existence of these integrals.

Lemma 1 For $i = 2, 3$, $j = 2, 3$ and $0 < u, v < 1$ we have

$$(i) \quad E(T_{n,i}(u)T_{n,j}(v)) \leq 4.5(u \wedge v - uv) \quad \text{for } n \geq 2$$

$$(ii) \quad E(T_{n,i}(u)T_{n,j}(v)) \rightarrow u \wedge v - uv$$

where $u \wedge v = \text{minimum of } u \text{ and } v$.

Proof First take $i = j = 2$ and let $X_n(u)$ and $X_n^*(v)$ be independent Binomial (n, u) and Binomial (n, v) random variables respectively. Note that $b_n(i, u) = \Pr(X_n(u) \geq i)$. From the formula for the covariance between two linear rank statistics (Hájek & Šidák (1967, page 62)) we have

$$\begin{aligned} E(T_{n,2}(u)T_{n,2}(v)) &= \left(\sum_{i=1}^n b_n(i, u)b_n(i, v) - nuv \right) / (n-1) \\ &= \left(\sum_{i=1}^n \Pr(X_n(u) \wedge X_n^*(v) \geq i) - nuv \right) / (n-1) \\ &= (E(X_n(u) \wedge X_n^*(v)) - nuv) / (n-1) \\ &= (E\{(X_n(u)/n) \wedge (X_n^*(v)/n)\} - uv)n / (n-1) \\ &\leq 2(u \wedge v - uv) \quad \text{for } n \geq 2. \end{aligned}$$

This proves (i). Also, by the (strong or weak) law of large numbers, $(X_n(u)/n) \wedge (X_n^*(v)/n) \rightarrow u \wedge v$. Then (ii) follows from the last equality above upon application of the bounded convergence theorem.

Next, take $i = j = 3$ and for $x > 0$ let $[x]$ = integer part of x .

Then

$$\begin{aligned} E(T_{n,3}(u)T_{n,3}(v)) &= ([(n+1)u] \wedge [(n+1)v] - [(n+1)u][(n+1)v]) / (n-1) \\ &\leq (u \wedge v - uv)(n+1)^2 / n(n-1) \\ &\leq 4.5(u \wedge v - uv) \quad \text{for } n \geq 2. \end{aligned}$$

The proofs for the cases $i \neq j$ are similar.

Lemma 2
$$\int_0^1 \int_0^1 (u \wedge v - uv) d\varphi(u) d\varphi(v) = A^2.$$

Proof By (6) and (10),

$$\begin{aligned}
 A^2 &= E\left(\int_0^1 T_{n,1}(v) d\varphi(v)\right)^2 \\
 &= \int_0^1 \int_0^1 E(T_{n,1}(u) T_{n,1}(v)) d\varphi(u) d\varphi(v) \\
 &= \int_0^1 \int_0^1 (u \wedge v - uv) d\varphi(u) d\varphi(v).
 \end{aligned}$$

See Hoeffding (1973, page 65) for another proof.

Proof of (7) and (8) From (12),

$$E(T_{n,2})^2 = \int_0^1 \int_0^1 E(T_{n,2}(u) \cdot T_{n,2}(v)) d\varphi(u) d\varphi(v) \rightarrow A^2$$

by Lemma 1, Lemma 2 and the dominated convergence theorem. The proof of (8) is similar.

REMARK

The integral representation (11) was used by Hájek (1968) in his proof of the asymptotic normality of linear rank statistics under general alternatives.

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